

Practical No. 18

Name : Danish Altaf Solkar

Class : CO - B

Enrolment no. : 24111590805

Batch : 2

1. Aim

To prepare an SQA plan for a software project to ensure that the developed product satisfies specified requirements and achieves important quality attributes such as correctness, reliability, usability, efficiency, maintainability, portability, and security.

2. Software Project

Project: Online Food Ordering System

The Online Food Ordering System allows customers to browse restaurants and food items, place orders, make payments, track orders, and provide feedback. The system also provides restaurant and administrator functions for managing menus, orders, users, and reports.

3. Quality Objectives

Quality Attribute	Quality Objective
Correctness	All specified functional requirements should be implemented correctly.
Reliability	The system should perform consistently without frequent failures.
Usability	The system should be simple, clear, and easy for users to operate.
Efficiency	The system should provide acceptable response time and resource usage.
Security	User data, passwords, payments, and access privileges should be protected.
Maintainability	The software should be easy to modify, debug, and enhance.
Portability	The application should work correctly on supported browsers and platforms.

4. SQA Activities

Sr. No.	SQA Activity	Description
1	Requirements Review	Review requirements for correctness, completeness, consistency, and testability.
2	SRS Review	Verify that the Software Requirements Specification accurately represents user needs.
3	Design Review	Check the software architecture, database design, interfaces, and modules.
4	Code Review	Inspect source code for coding standards, errors, security issues, and maintainability.
5	Testing	Perform unit, integration, system, usability, security, and regression testing.
6	Defect Management	Record, classify, track, fix, and verify software defects.
7	Configuration Management	Control versions of source code, documents, test cases, and releases.
8	Quality Audits	Check whether project processes and standards are being followed.
9	Metrics Collection	Collect defect density, test coverage, failure rate, and other quality measures.
10	Final Quality Review	Verify that all quality objectives and acceptance criteria are satisfied before release.

5. Quality Standards and Guidelines

- Follow approved software development and coding standards.
- Use a standard format for requirements, design documents, source code, test cases, and defect reports.

- Apply secure coding practices for authentication, authorization, input validation, and data protection.
- Maintain version control for all important project artifacts.
- Perform reviews at each major stage of the software development life cycle.

6. Testing Plan

Testing Type	Purpose
Unit Testing	Test individual modules or functions independently.
Integration Testing	Verify communication and data flow between integrated modules.
System Testing	Test the complete system against specified requirements.
Usability Testing	Check whether users can operate the system easily and effectively.
Security Testing	Identify authentication, authorization, input validation, and data protection issues.
Performance Testing	Measure response time, throughput, and system behavior under expected load.
Regression Testing	Ensure that new changes do not break previously working features.
Acceptance Testing	Confirm that the system meets customer and business requirements.

7. Defect Management Process

- Identify and report the defect.
- Assign a unique defect ID and record the defect details.
- Classify the defect according to severity and priority.
- Assign the defect to the responsible developer.
- Fix the defect and update its status.
- Retest the corrected functionality.
- Perform regression testing where required.
- Close the defect after successful verification.

8. Quality Metrics

Metric	Purpose
Defect Density	Measures the number of defects per unit size of software.
Test Case Pass Percentage	Measures the percentage of test cases that pass successfully.
Test Coverage	Measures how much of the software functionality is covered by testing.
Defect Removal Efficiency	Measures the effectiveness of defect detection and removal.
Failure Rate	Measures how frequently the software fails during operation.
Mean Time Between Failures	Measures the average operating time between failures.

9. Roles and Responsibilities

Role	Responsibility
Project Manager	Plan and monitor quality activities and ensure compliance with the SQA plan.
SQA Engineer	Prepare quality procedures, conduct audits, and monitor quality metrics.
Developer	Follow coding standards, perform code reviews, and correct defects.
Tester	Prepare test cases, execute tests, report defects, and perform retesting.
Customer/User	Provide feedback and participate in acceptance testing.

10. SQA Schedule

Project Phase	SQA Activity
Requirement Analysis	Requirements review and SRS verification
Design	Architecture and database design review
Implementation	Code review and coding standard verification
Testing	Test planning, test execution, defect tracking, and regression testing
Release	Final quality audit and acceptance testing
Maintenance	Change review, regression testing, and defect monitoring

11. Expected Result

The SQA plan provides a systematic approach for assuring software quality. It helps identify defects early, ensures compliance with defined standards, monitors quality attributes, and verifies that the final Online Food Ordering System is reliable, secure, usable, efficient, maintainable, and suitable for its intended purpose.

12. Conclusion

An effective Software Quality Assurance plan ensures that quality is considered throughout the software development life cycle rather than only during final testing. By applying reviews, testing, audits, defect management, configuration management, and quality metrics, the development team can deliver a quality software product that satisfies user requirements.